Marmara University Electrical-Electronics Eng. Dept.

CSE2003 Digital Design Laboratory Assignment #2

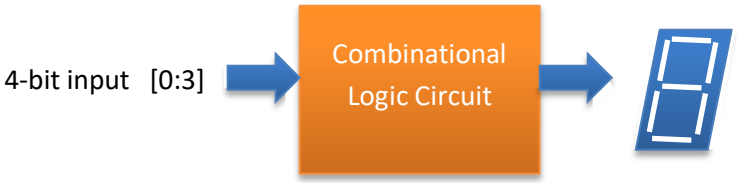
The studies will be uploaded to the system before next lab section. **Each group must present their own study. The group members have to work together and each member must know each step of the study.**

For this laboratory assignment, you will use Altera Quartus II software and DE-115 FPGA kit and design a binary to hexadecimal converter that has the Input/output defiitions below.

The steps you need to follow: 1. You should design the circuit, **2.**write the verilog HDL code fort he designed circuit, **3.** Check if the designed hardware is correctly coded by using RTL viewer. Don't forget: your circuit may look different then yours since the compiler can make some optimizations. Please take this information into account when you compare the RTL viewer circuit and yours. **4.** Verify your design by using simulation tool (you need to check the inputs and outputs). **5.** implement and RUN your code using Altera DE-115 FPGA kit. Use the remote desktop connection (**Remote desktop connection ID**:256 521 711 **password**: dijitallab2021).

You need to upload the following files before due date:

1. A short video (<60 secs.) that shows all steps of the study. Especially, the implementation part must be included.
2. The report (in PDF format): 1. What is the purpose of the study? 2. What are the design details with necessary explenations. 3. The design and implementation steps (all steps with screenshots etc.). 4. The result and coclusion: did you reach the purposed result? If the answer is no, why? if it is yes, how?

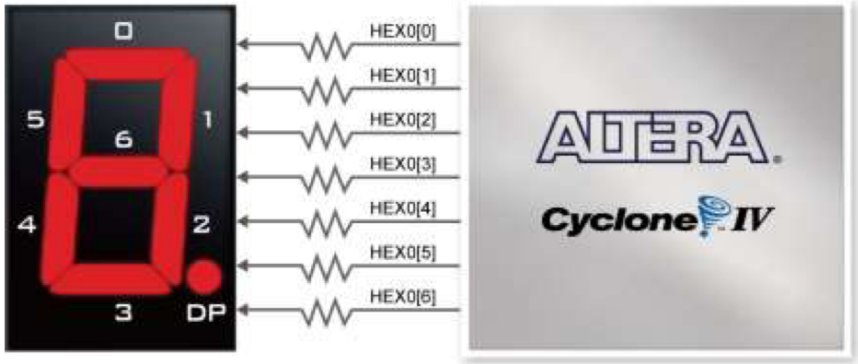


Hexadecimal (base 16) is a common way to represent numbers in digital circuits. Each hex digit shows the value of 4 bits of the number (value 0-15):

Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Representation in hex	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Your task is to implement a circuit that displays the value of a 4-bit number as a hex digit(use the 'sw' inputs). The hex digit will be displayed on the 7-segment.

You can find the connection diagram and FPGA pin tables below:



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Signal Name	*FPGA Pin No.*	*Description*
HEX0[0]	PIN_G18	Seven Segment Digit 0[0]
HEX0[1]	PIN_F22	Seven Segment Digit 0[1]
HEX0[2]	PIN_E17	Seven Segment Digit 0[2]
HEX0[3]	PIN_L26	Seven Segment Digit 0[3]
HEX0[4]	PIN_L25	Seven Segment Digit 0[4]
HEX0[5]	PIN_J22	Seven Segment Digit 0[5]
HEX0[6]	PIN_H22	Seven Segment Digit 0[6]

SW1->AC15

SW2->Y17

SW3->Y16

SW4->AE15

SW5->AF16

SW6->AF15

SW7->AE21

SW8->AC22

SW9->AF21

SW10->AD22

SW11->AD25

SW12->AE25

SW13->AE24

SW14->AF26

SW15->AG23

SW16->AB22

SW17->AB21

SW18->AC21

SW19->AD21

PB->AD15